Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year 2021-2022
Semester Fall
Exam Date 28 /11/ 2021

QUESTIONS FOR THE MID TERM EXAM

Subject: (ELC 422) Digital Signal Processing
Examiners: Dr. Samir Kamal & Dr. Sara Hassan

Spec.: Level- 4, Elec. Eng. & Comm.Tech. Dep.
Time: 60 Min

Number of Pages: one | Number of Questions: 2 | Attempt all questions

Model - B

Question 1 [20 Points]

a- In A/D, in detail, explain the sampling process of analog signals.

b- Consider the following analog sinusoid signal: $x(t) = 3\sin(100\pi t)$

(i) Sketch the signal for [0, 30] ms.

(ii) The signal $x(t)$ is sampled with sampling rate F_S=300 samples per second. Determine the frequency of the discrete-time signal X(n) and show that it is periodic.

(iii) Compute the sample values in one period of X (n). Sketch X (n) on the same diagram with $x(t)$.

Question 2 [20 Points]

a- Prove that the DFT, X(k), of a sequence x(n) is periodic

b- For the following sequence, x (n): x (n) = {1, 0, 1, 0}:

(i) Using the DFT method find the frequency components of x(n), X(k)

(ii) Using X(k) obtained in part (i), find x(n) using the IFFT method.

(iii) Using your results, verify Parseval's theorem.

(iv) Plot the amplitude and the phase spectrums of X(k).

GOOD LUCK